

TITLE PAGE

- Problem Statement ID – SIH26013
- Problem Statement Title – Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management
- Theme – Disaster Management
- PS Category- Software
- Team ID-
- Team Name - KOSHIN



Idea

- ✓ Geo Harmon-X is an AI-powered platform that integrates cadastral maps, satellite images, revenue records, municipal GIS, utility maps, and survey data into one trusted parcel-level land record.

Detailed Explanation

- ✓ Collects data from multiple geospatial sources.
- ✓ Detects different CRS, formats, and schemas automatically.
- ✓ Resolves conflicts using confidence scoring and validates geometry before creating a master record.

Addressing the Problem

- ✓ Urban land records are scattered across different departments and often contain inconsistent parcel information.
- ✓ Geo Harmon-X harmonizes these datasets, removes duplicates, resolves conflicts, and improves the accuracy and transparency of land records.

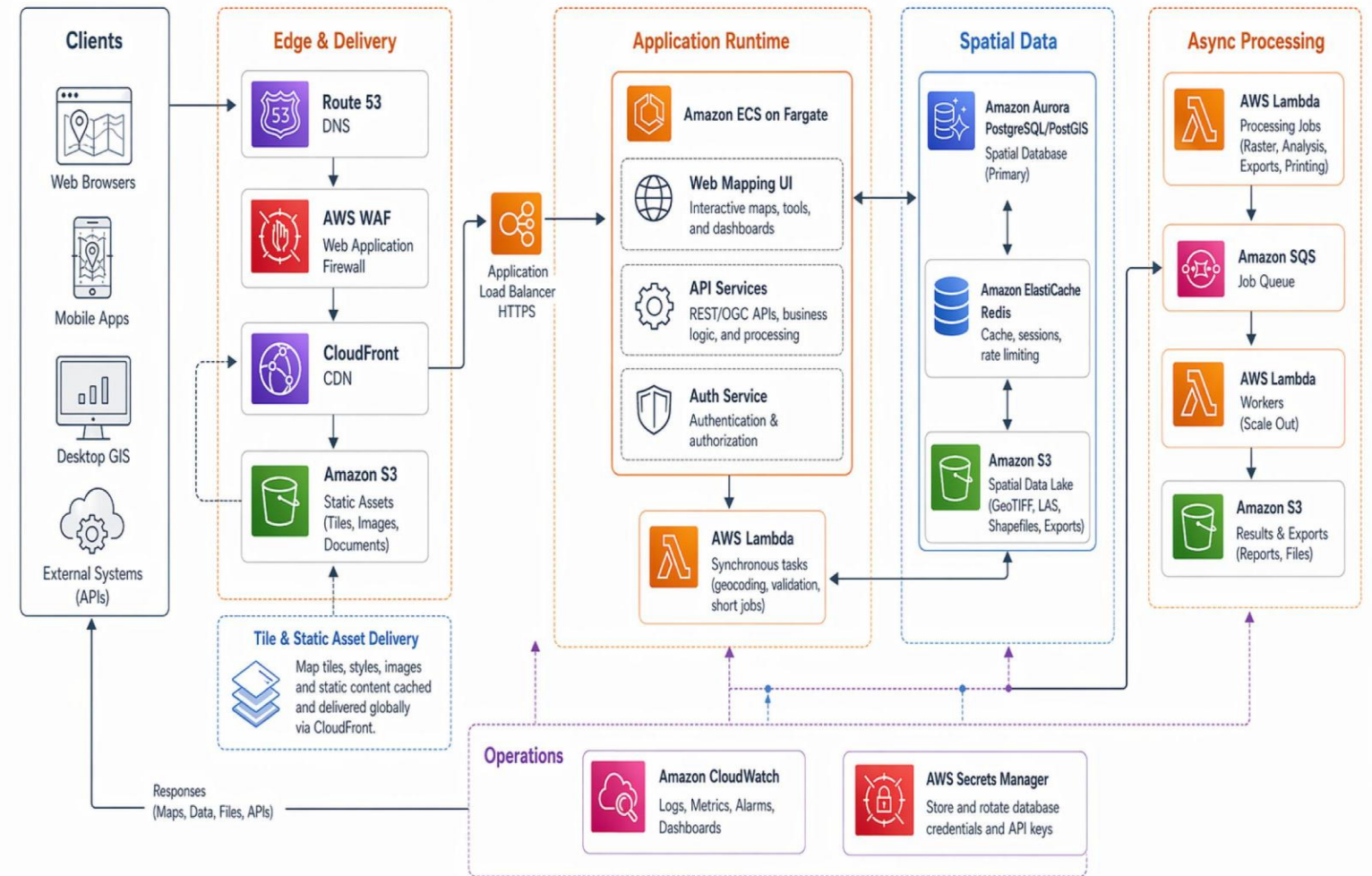
Innovation / Uniqueness

- ✓ AI-based parcel matching.
- ✓ Confidence-weighted conflict resolution.
- ✓ Human-in-the-loop validation.
- ✓ Version history and provenance for every land record.

Workflow

1. Collect multi-source geospatial data.
2. Standardize coordinate systems.
3. AI aligns and merges datasets.
4. Detect overlapping or missing boundaries.
5. Generate unified GIS database.
6. Display results on Web GIS dashboard.

- ** Python + FastAPI
- ** QGIS / GeoServer
- ** PostGIS (Spatial Database)
- ** Machine Learning + Computer Vision



Challenges



Different CRS



Missing parcel IDs



Conflicting Attributes



Invalid Geometry



Duplicate Parcels



Uncertain AI Predictions



Large Datasets



Strategy



Automatically detect and transform CRS, then validate residual errors.



Use spatial and semantic entity resolution to identify matching parcels.



Use confidence-weighted source ranking to select the best-supported value.



Apply topology validation and automated geometry repair.



Use spatial similarity and AI-based candidate matching.



Send low-confidence cases for human-in-the-loop validation.



Use spatial indexes, tiled processing and asynchronous ETL.

Potential Impact

•Government:

- **Faster Land Records:** Reduces manual GIS integration work.
- **Better Accuracy:** Creates consistent and validated parcel information.

•Urban planning:

- **Better Planning:** Provides reliable parcel data for zoning and infrastructure.
- **Faster Decisions:** Makes different GIS datasets available in one system.

•Revenue Administration:

- **Accurate Parcels:** Improves parcel identification.
- **Consistent Data:** Helps maintain reliable assessment information.

•Citizens:

- **Economic Growth:** More income from better crop yields.

Benefits

•Social:

- **Transparency:** Citizens can rely on more consistent land information.
- **Efficient Governance:** Departments can work with harmonized data.

•Economic:

- **Reduced Costs:** Less manual GIS processing and data correction.
- **Faster Processing:** Automated matching reduces repetitive work.

•Technical:

- **Interoperability:** Combines data from different GIS sources
- **AI + GIS:** AI handles uncertain parcel matching while GIS rules validate spatial correctness.
- **Scalability:** Spatial indexing and tiled processing support large datasets.

- Details / Links of the reference and research work
- <https://www.ogc.org/groups/land-administration-domain-working-group/>
- ISO 19152-1:2024 – Land Administration Domain Model (LADM)
- <https://www.iso.org/standard/81263.html>